

Dispersion in $\text{GeO}_2\text{-SiO}_2$ glasses

James W. Fleming

Germania glass was prepared from high purity GeO_2 powder. Refractive-index dispersion was used to calculate material dispersion and to provide a model for representing the dispersion of $\text{GeO}_2\text{-SiO}_2$ glasses. The wavelength of zero material dispersion is found to be in agreement with theoretical calculations. Modal propagation is modeled for a GeO_2 core-silica clad fiber. Results support compositional dependence of profile dispersion in $\text{GeO}_2\text{-SiO}_2$ fibers.

I. Introduction

The $\text{GeO}_2\text{-SiO}_2$ glass system has important applications in fiber optics and optical communications. The high silica end of the binary (<20 mole % GeO_2) is currently of greatest interest. However, interest is increasing in the high germania side for the following reasons. As silica, germania is an excellent glass former. Few other oxides form glasses as readily. Additionally, previous calculations have shown GeO_2 glass might exhibit a lower optical loss window than that of high SiO_2 containing glasses currently used for lightguides as a result of reduced Rayleigh scattering at longer wavelengths.¹⁻⁴ Supporting its use as a lightguiding medium, the point of zero material dispersion was calculated to fall within the low loss window.

For the purpose of understanding the optical properties of $\text{GeO}_2\text{-SiO}_2$ glasses throughout the complete system a model of refractive-index dispersion based on a three-term Sellmeier dispersion relation is reported. Little is known about the refractive-index dispersion of glass containing >20 mole % GeO_2 due to the difficulty in fabricating suitable specimens of good optical quality. The model can be used for accurate calculations of the dispersive properties of an entire material system if the characteristics of the end members are known. It is suggested as a replacement for the linear dependence of index on composition which is often employed in refractive-index calculations.

Another goal in the model construction is to discern the nature of profile dispersion in this system and to assess its importance to the bandwidth of $\text{GeO}_2\text{-SiO}_2$ based lightguides. A power law relationship is com-

monly used to describe the optimum profile shape for minimizing intermodal dispersion in multimode optical fibers. This relationship is correct only if the refractive index and the first derivative of refractive index are both linearly dependent on composition. If this assumption is incorrect, the profile which optimizes bandwidth will not be a power law relationship. The model for refractive-index dispersion can be used to discern the profile dispersion and the possible error resulting from the above linearity assumption.

II. Experimental

The germania glass was prepared by melting 99.9999% pure hexagonal GeO_2 powder in a Pt.-foil-lined alumina crucible. Melting was done at 1550°C in an electric resistance furnace open to the atmosphere. The material was held at this temperature for 2 h, which was sufficient to permit fining of the melted batch. From the melting furnace it was transferred to an annealing furnace preheated to 700°C . The sample was soaked at 700°C for 1 h and then cooled to room temperature at 20°C/h . The resultant glass has good homogeneity. A prism was fabricated from a piece of this sample having a prism angle of 30° and prism faces polished to $1/10$ wave flatness. Another piece was polished for optical attenuation measurement.

The indices of the GeO_2 prism were determined at 38 wavelengths from 0.36 to $4.3\ \mu\text{m}$. An automated refractive-index measurement facility, employing the autocollimation technique, with a precision of 2×10^{-5} was used for the measurements.⁵ The temperature of the prism during measurement varied from 22 to 26°C . Two or three measurements were made at each wavelength on different days. The same sample was used for all measurements and was kept in a dry N_2 environment to avoid moisture attack throughout the experiment. No such attack was observed. The standard deviation of the measurements is $<4 \times 10^{-5}$. Index measurements were fit to a three-term Sellmeier dispersion relation of the form

The author is with AT&T Bell Laboratories, Murray Hill, New Jersey 07974.

Received 26 July 1984.

0003-6935/84/244486-08\$02.00/0.

© 1984 Optical Society of America.

Table I. Average Values of Measured Refractive Indices

$\lambda(\mu\text{m})$	n
0.365015	1.64605
0.404656	1.63393
0.435835	1.62697
0.467815	1.62137
0.479991	1.61956
0.508582	1.61585
0.546074	1.61192
0.578012	1.60918
0.589262	1.60832
0.643847	1.60481
0.852110	1.59692
0.894350	1.59591
0.920846	1.59529
1.002439	1.59377
1.012360	1.59354
1.01398	1.59356
1.03946	1.59313
1.12866	1.59180
1.19332	1.59095
1.3589	1.58905
1.3622	1.58902
1.4695	1.58794
1.52958	1.58731
1.693	1.58579
1.6932	1.58567
1.81307	1.58460
1.859	1.58415
1.97009	1.58306
2.325347	1.57936
2.373	1.57895
2.717	1.57493
2.764	1.57433
4.2369	1.54955
4.2809	1.54838

$$n^2 - 1 = \sum_{i=1}^3 \frac{A_i \lambda^2}{\lambda^2 - l_i^2}, \quad (1)$$

where n is the refractive index,
 A_i is the oscillator strength,
 l_i is the oscillator wavelength, and
 λ is the wavelength.

The fitting routine used is NLLSQ.⁶ The average absolute value of the residuals is $<2 \times 10^{-5}$. A standard F test of the residual indicates that there is no reason to doubt that the formula accurately represents the data.

Optical attenuation of a 1-cm thick specimen was measured from 0.2 to 2.5 μm using a Carey 14 spectrophotometer and from 2.5 to 6.0 μm using a Perkin-Elmer model 283B spectrometer.

III. Results and Discussion

A. Refractive-Index Dispersion

Average refractive indices are given in Table I for different wavelengths. The temperature dependence of the dispersion has not been determined so that no correction could be made for index variation with temperature during measurement. Such deviation is part of the standard error of 4×10^{-5} . The coefficients resulting from the fit of Eq. (1) are given in Table II. The model accurately represents the data over the complete wavelength range measured.

The refractive index at $\lambda = 0.5893$, density, and thermal expansion coefficient have been previously determined for other pure GeO_2 glasses.⁷⁻¹¹ The data obtained on this sample including thermal expansion and density agree with these other results to within one standard deviation of their respective measurements. Considering the typical variation of the refractive index of glasses with thermal history, the variation here is quite acceptable. It is expected that variations in GeO_2 refractive indices will be primarily due to thermal history.

From an analytic determination of $d^2n/d\lambda^2$ from Eq. (1), the point of zero material dispersion is found to be 1.738 μm . This is a good agreement with recent theo-

Table II. Sellmeier Coefficients for SiO_2 from Malitson¹⁹ and for GeO_2 from this Work

A_1	l_1	A_2	l_2	A_3	l_3
SiO_2					
0.69616630E 00	0.68404300E-01	0.40794260E 00	0.11624140E 00	0.89747940E 00	0.98961610E 01
GeO_2					
0.80686642E 00	0.68972606E-01	0.71815848E 00	0.15396605E 00	0.85416831E 00	0.11841931E 02

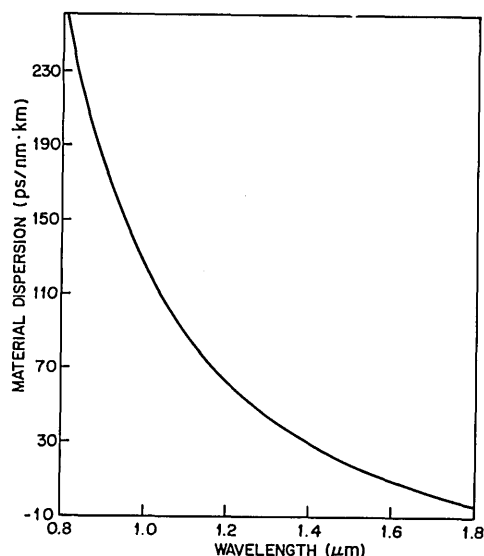


Fig. 1. Material dispersion for GeO₂ glass from 0.8 to 1.8 μm. Zero crossing is 1.738 μm.

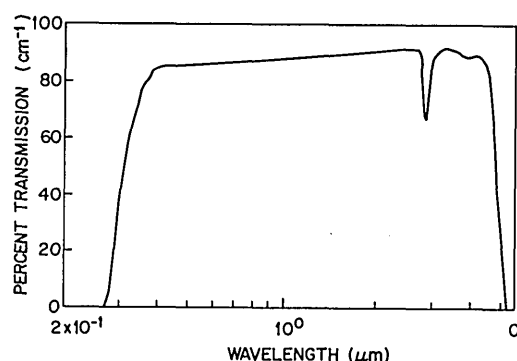


Fig. 2. Transmission of a 1-cm thick piece of GeO₂ glass.

retical calculations^{1,2,4} and in excellent agreement with recent measurements by other investigators.¹¹ Figure 1 shows the material dispersion as a function of λ for this GeO₂ glass.

While this specimen gives no indication of the optimum loss obtainable with GeO₂ glass, absorption measurements show that the intrinsic IR absorption is far removed from the region of zero material dispersion. Figure 2 shows the transmission characteristics of this material. The apparent variation in loss in the transparent region is an artifact of the sample surface quality.

The Sellmeier parameters for GeO₂ obtained from the fit provide an opportunity for modeling the refractive-index dispersion of all GeO₂-SiO₂ compositions. These parameters represent characteristics of the intrinsic absorption bands of the material, and it has been shown that a three-term Sellmeier equation with two oscillators in the UV and one in the IR provides a very precise representation of refractive-index data. Since there is a physical link between the parameters and structural characteristics of the materials, one can assume that as the composition of a glass is changed by the addition of a second component, the Sellmeier coefficients are going to shift in a continuous fashion. It is proposed that such

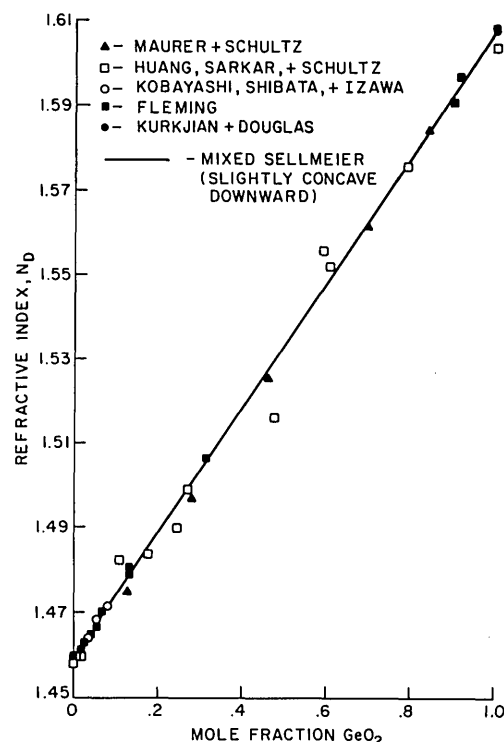


Fig. 3. Refractive index at $\lambda = 0.5893$ vs composition for GeO₂-SiO₂ glasses prepared and measured by various researchers. The line represents the computed values using the mixed Sellmeier model.

is the case for the GeO₂-SiO₂ system and that the variation of any parameter from the best fit value for SiO₂ to that for GeO₂ can be approximated by a simple linear relationship involving the respective parameters of the end members. The relationship is dependent only on composition and can be represented as follows:

$$n^2 - 1 = \sum_{i=1}^3 \frac{[SA_i + X(GA_i - SA_i)]\lambda^2}{\lambda^2 - [Sl_i + X(Gl_i - Sl_i)]^2}, \quad (2)$$

where SA , Sl , GA , Gl are the Sellmeier coefficients for the SiO₂ and GeO₂ glasses, respectively, and X is the mole fraction of GeO₂. This general type of model has been used by others^{12,13} on various binary glass systems. In those studies agreement with predicted and empirically determined index values was found to be good.

Equation (2) has been used to determine n vs composition at 0.5893 μm for which most of the index data on GeO₂-SiO₂ glasses have been measured. Figure 3 shows a comparison of measured refractive-index values available in the literature^{5,9-11,14-17} and those calculated from the model. In Fig. 4 the variation is shown for the composition range currently most often employed in lightguide fabrication. Included in Fig. 4 are measurements made on a fiber having several steps of different GeO₂ concentration.¹⁸

The parameters used in Eq. (2) were those obtained by Malitson¹⁹ for SiO₂ glass and those obtained in this work for GeO₂. It can be seen that the mixed Sellmeier model adequately represents the data. A linear relationship would work apparently as well. However, the mixed representation has the obvious advantage of

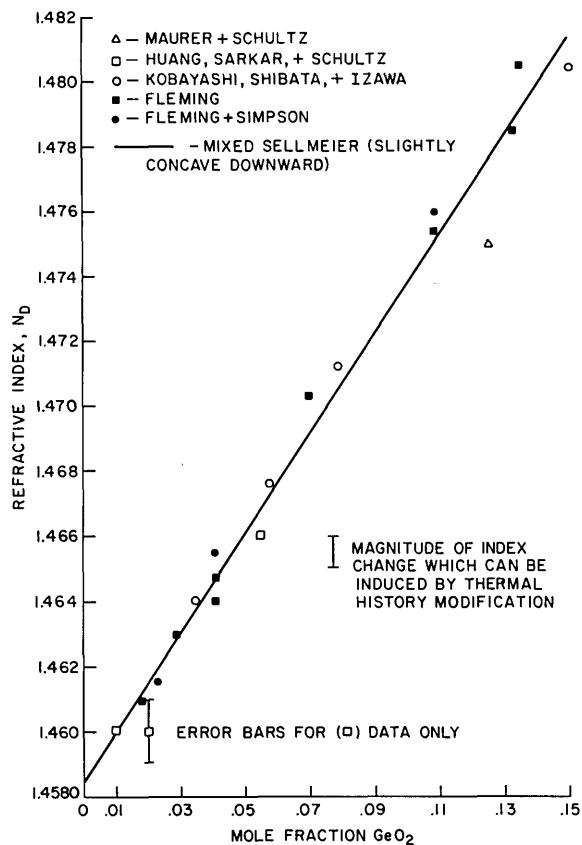


Fig. 4. Portion of Fig. 3 showing composition region of current interest for lightguides.

being useful for determining the complete dispersion curve for any composition of interest and prediction of the other properties which can be derived therefrom. Such data were previously available on only those compositions near the silica side of the binary, and few results have been duplicated by different researchers on the same compositions to give a feeling for the anticipated variation as a function of material source and preparation technique.

To show the quality of the fit at another wavelength the data for index from Refs. 5, 14, and 17 obtained at $1.3622 \mu\text{m}$ have been plotted in Figs. 5 and 6 along with the indices determined from Eq. (2). Again it can be seen that the representation is good. An important observation of the $1.3622\text{-}\mu\text{m}$ data is that Eq. (2) yields less of a straight line at longer wavelengths. This is particularly true in the wavelength range of anomalous dispersion around the zero material dispersion wavelength.

The samples from which the parameters of the Sellmeier equation were obtained were annealed glasses. As shown in Fig. 4 the variation in index as a function of thermal history can be as large as 1×10^{-3} .¹⁶ This thermal history dependence should be included in any precise model of refractive-index dispersion in glass. It too enters the model through some effect on the parameter values. Exactly how has not yet been determined. However, it is found that $dn/d\lambda$ does not vary

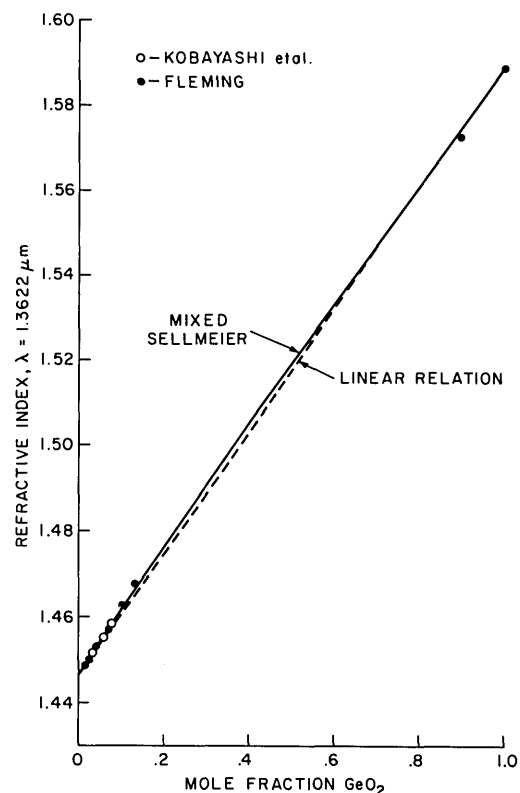


Fig. 5. Refractive index at $1.3622 \mu\text{m}$ vs composition for $\text{GeO}_2\text{-SiO}_2$ glasses prepared and measured by two researchers.

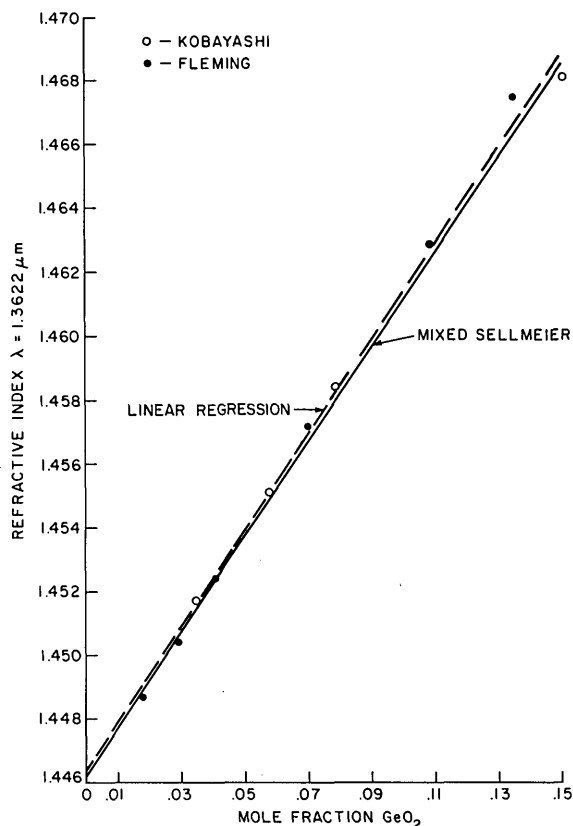


Fig. 6. Portion of Fig. 5 showing composition region of current interest for lightguides.

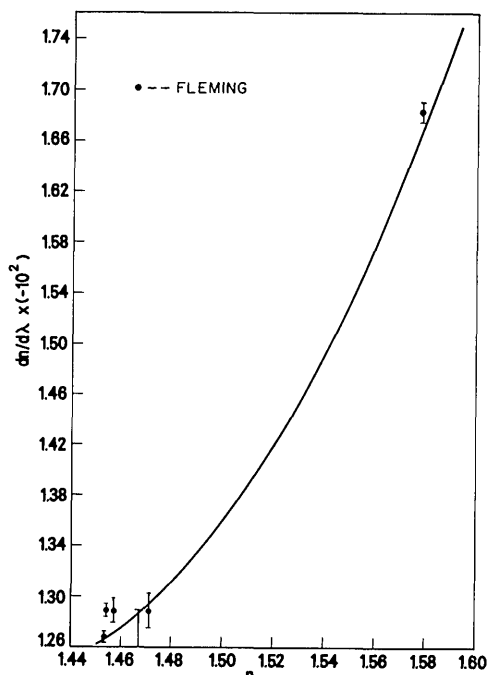


Fig. 7. Dependence of $dn/d\lambda$ on n at $1.0 \mu\text{m}$ for $\text{GeO}_2\text{-SiO}_2$ glasses computed from the mixed Sellmeier model.

significantly with thermal history,¹⁷ and the resultant change as a function of composition for small composition variations is small. Therefore, it is sufficient for calculations of Δn in systems of moderate N.A. (0.15–0.25) to use the annealed values.

For a specific range of compositions where the amounts of only two components are varying, the same model can be applied with greater accuracy. For example, given the parameters for SiO_2 and $0.135\text{GeO}_2 \cdot 0.865\text{SiO}_2$ glasses a model could be obtained which provides dispersion estimates for glasses in the encompassed composition range.

B. Modal Dispersion

To minimize modal dispersion in multimode lightguides the refractive-index profile of the core is tailored so that the various modes of light traversing a length of fiber do so at the same rate. Modal dispersion is dependent on the first derivative of the refractive index with respect to wavelength for the material at any point in the lightguiding part of the fiber. With the correct choice of materials comprising this optically active portion of the fiber, the intermodal dispersion can be greatly reduced. This material choice is based on n and $dn/d\lambda$. The objective is a fiber in which the refractive-index profile has a specific shape and specific variation in shape with wavelength.

The radial index profile can be described by

$$n^2(r) = n_1^2[1 - 2\Delta \cdot f(r,a)], \quad r < a, \quad (3)$$

where $f(r,a)$ is the profile function, r is the distance from the fiber axis, a is the core radius, and $\Delta = (n_1 - n_2)/n_1$, where n_1 is the axial core refractive index, and n_2 is the cladding refractive index. This is a general refractive-index-profile equation. When f is $(r/a)^\alpha$ we obtain

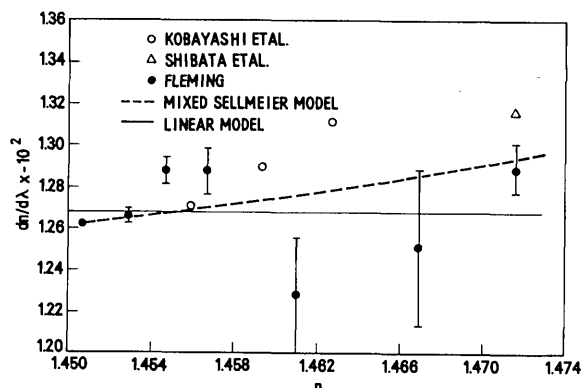


Fig. 8. First derivative of refractive index vs index at $1.0 \mu\text{m}$ showing empirically obtained points of Fleming, Kobayashi, and Shibata and the computed line from the mixed Sellmeier model.

the power-law-profile equation used most often to characterize profile shapes in lightguides.

For the special case where n and $dn/d\lambda$ are linearly related, the profile shape remains power law with λ . The change in profile shape is called profile dispersion. An assumption of constant profile dispersion was first introduced by Olshansky and Keck.²⁰ It greatly simplifies the calculation of the optimum profile shape for minimizing modal dispersion. It was assumed that profile dispersion would change for different glass systems but not for different compositions within a given system. Of particular importance is the nature of profile dispersion in the $\text{GeO}_2\text{-SiO}_2$ glasses since GeO_2 and SiO_2 are the major components being used. Previous dispersion measurements on bulk glasses indicate that there will be a nonpower law change in profile shape with a wavelength in fibers having $\text{GeO}_2\text{-SiO}_2$ cores^{14,17,21} because a nonlinearity is evident in n and $dn/d\lambda$. Measurements on fiber specimens do not show this.^{22,23}

There is no theory in the physics of light propagation in solids, which suggests that a linear relationship between n and $dn/d\lambda$ should exist in even a simple glass system. A model such as Eq. (2) has more theoretical justification because it involves the characteristics of the material responsible for dispersion. However, the variation of $dn/d\lambda$ with n as a function of changing composition may be sufficiently close to linear that this becomes a good approximation. This has appeared to be the case when dealing with small dopant additions such as 0–10 mole % of GeO_2 in SiO_2 and has lead to high bandwidth multimode fibers.

One can readily see that Eq. (2) will not provide such a linear relationship. As shown in Fig. 7, n and $dn/d\lambda$ are clearly not linearly related for $\text{GeO}_2\text{-SiO}_2$ glasses at $\lambda = 1.0 \mu\text{m}$. Similar behavior is observed at other wavelengths. However, as shown in Fig. 8 the data generated in the wavelength range of current interest for optical transmission are approximately linearly related. Also plotted is data for bulk glasses accompanied where available by error bars indicating the expected standard deviation for these measurements. The standard deviation was determined by computing the

variance of a Taylor series approximation of the derivatives of Eq. (1) using calculated covariances for the fitted parameters determined from the fitting program.²⁴

If it is assumed that Eq. (2) accurately models the dispersion behavior in the $\text{GeO}_2\text{-SiO}_2$ glass systems, there exists a relationship between n and $dn/d\lambda$, which appears to closely approximate a linear relationship between the above two quantities. Indeed, as shown in Figs. 3–6, the deviation from a linear relationship between n and concentration is from the figures almost undetectable. Therefore, the optimum profile shape should be the same for all waveguides comprised of any core-cladding glass combination in the $\text{GeO}_2\text{-SiO}_2$ glass system as has been previously predicted.²⁰ However, considering the extremes of a GeO_2 core and a SiO_2 cladding, these glasses have such markedly different dispersive characteristics that it is not expected that a lightguide comprised of them will have the same optimum profile shape as, for example, a lightguide with a $0.10\text{GeO}_2 \cdot 0.90\text{SiO}_2$ core and an SiO_2 cladding. Figure 9 shows a comparison of the optimum α vs λ characteristics of a GeO_2 core lightguide computed from the bulk data and that for an approximately $0.13\text{GeO}_2 \cdot 0.87\text{SiO}_2$ core SiO_2 clad lightguide as found by several investigators, including calculations from measurements of lightguides^{22,23} and bulk glasses.²⁵ The optimum α was calculated in all cases using the Olshansky and Keck approximation based on WKB method²⁰:

$$\alpha = 2 + P - \Delta \frac{(4 + P)(3 + P)}{(5 + 2P)}, \quad (4)$$

where

$$P = \frac{-2n_1 \lambda}{N_1} \frac{d\Delta}{d\lambda}, \quad (5)$$

$$\Delta = \frac{n_1 - n_2}{n_1}, \quad (6)$$

n_1 = axial core phase index,
 n_2 = cladding phase index, and
 N_1 = core group index.

The quantity P is the profile dispersion parameter. For the assumption of a constant profile dispersion it is a constant. The data in this figure clearly indicate that profile dispersion varies in fibers comprised strictly of $\text{GeO}_2\text{-SiO}_2$ glasses. Another significant feature of the data in Fig. 9 is that there is good agreement between the bulk and fiber determinations within this wavelength range for the 0.23 N.A. fibers. Considering the reported variation in the calculation (± 0.02),^{22,23} at a low confidence level, there is insufficient evidence to reject a hypothesis that the curves are the same at any wavelength in this range regardless of the source of data or method of calculation.

The effect on profile dispersion of a slight deviation from linearity in n and $dn/d\lambda$ can be examined using Eq. (2). The optimum α was determined for the following lightguides using calculations from Eqs. (2) and (4): a 0.2 N.A. fiber having a $0.09\text{GeO}_2 \cdot 0.91\text{SiO}_2$ core and an SiO_2 cladding, a 0.3 N.A. fiber having a $0.20\text{GeO}_2 \cdot 0.80\text{SiO}_2$ core and a SiO_2 cladding, a 0.2 N.A. fiber

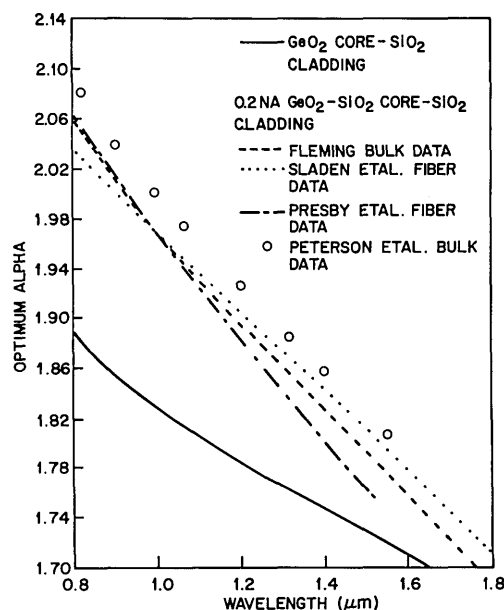


Fig. 9. Optimum α as a function of wavelength for fiber and bulk calculations of 0.23 N.A. $\text{GeO}_2\text{-SiO}_2$ core- SiO_2 clad lightguides and a GeO_2 core- SiO_2 clad lightguide.

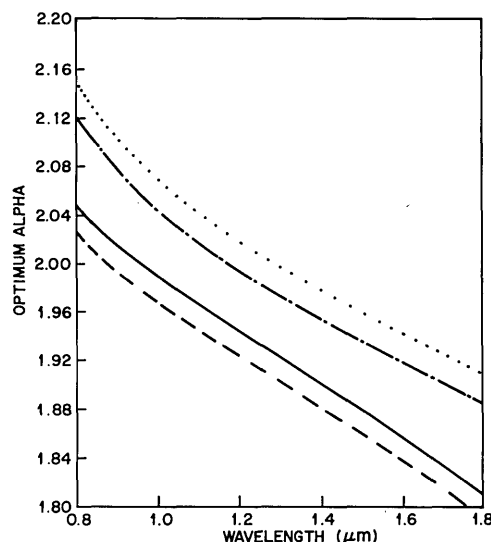


Fig. 10. Optimum α as a function of wavelength using the mixed Sellmeier model for a 0.2 N.A. lightguide having a $0.09\text{GeO}_2 \cdot 0.91\text{SiO}_2$ core and an SiO_2 cladding —, a 0.02 N.A. fiber having a GeO_2 core and a $0.91\text{GeO}_2 \cdot 0.09\text{SiO}_2$ cladding ···, a 0.3 N.A. fiber having a $0.20\text{GeO}_2 \cdot 0.80\text{SiO}_2$ core and a SiO_2 cladding —, and a 0.3 N.A. fiber having a GeO_2 core and a $0.81\text{GeO}_2 \cdot 0.19\text{SiO}_2$ cladding — ·—.

having a GeO_2 core and a $0.91\text{GeO}_2 \cdot 0.09\text{SiO}_2$ cladding, and a 0.3 N.A. fiber having a GeO_2 core and a $0.81\text{GeO}_2 \cdot 0.19\text{SiO}_2$ cladding. The results appear in Fig. 10. As can be seen slight deviations from the linearity assumption can cause variations in the optimum profile shape. Again, since there is no physical reason to predict a truly linear relationship between n and $dn/d\lambda$, the assumption is possibly erroneous and could affect the accuracy of an optimum profile shape calculated using a relationship based on that assumption.

One can get an indication of the relative magnitude of the error in calculation by comparing the optimum profile data obtained from an exact numerical solution to Maxwell's equations such as that of Peterson *et al.* with that using approximation methods such as employed above.²⁶ For convenience, the exact numerical solution results are also shown in Fig. 9 for a fiber having a $0.135\text{GeO}_2 \cdot 0.865\text{SiO}_2$ core and a SiO_2 cladding. In this case the difference in optimum α is ~ 0.03 . While this is a significant variation with regard to modal dispersion, it is not large from the perspective of current manufacturing and profile measurement methods.

Examining the bulk data given in Fig. 8, linear regression of $dn/d\lambda$ on n suggests a very poor linear dependence as does the graph. Analysis of variance on the data of this author employing the F test indicates that there is reason to reject the hypothesis that the data are linearly related. The confidence level is $>90\%$ ($P < 10\%$). A normal distribution and constant variance are assumed for the population.

Also appearing on Fig. 8 are computed values obtained from the work of Kobayashi *et al.*¹⁴ and Shibata *et al.*²⁷ These measurements on glasses made by VAD are in the same range as those obtained in the present study. Insufficient data were presented in the respective publications to permit a calculation of standard deviation. However, it is reported in Ref. 14 that the standard deviation in the index measurement was 1×10^{-4} . The reason for this larger deviation seems to be the result of composition fluctuations inherent to the glass fabrication technique. These composition changes caused index variations approaching 5×10^{-4} . The quality of the fitting routine used to fit the model to their data must also be considered. Without measurements of the standard deviation a determination of the applicability of a linear model to the VAD specimen data cannot be made. However, the published values plotted in Fig. 8 do not support a linear model considered separately or jointly with the other measurements. Experimental evidence on bulk samples appears sufficient to reject the assumption of constant profile dispersion.

The observed compositional change in refractive-index dispersion may be a result of complex structural variations within the specimens which are dependent on composition and fabrication technique. There is an interesting correlation between the Raman spectra of the $\text{GeO}_2\text{-SiO}_2$ glasses and the observed dependence of $dn/d\lambda$ on n . Examination of Fig. 8 indicates the inflection between the index values for 4.1 and 7.1 mole % GeO_2 . This appears to be the compositional region of most pronounced fluctuation regardless of wavelength. Sproson *et al.*²⁸ found that a new peak was observed in this composition range and that another discontinued its growth. Perhaps these structural changes are largely responsible for the nonlinearity of $dn/d\lambda$ and n in this composition range.

The bulk data indicate a greater nonlinearity in n and $dn/d\lambda$ than does the model. To evaluate the effect this will have on profile shape it is possible to calculate the magnitude and direction of the change in profile shape with wavelength which would result from the dispersion

indicated by the bulk sample measurements. Using the power-law-profile equation the refractive index as a function of core radius was computed for a fiber having a $50\text{-}\mu\text{m}$ o.d. core, a $13.5\text{GeO}_2 \cdot 86.5\text{SiO}_2$ axial composition, a SiO_2 cladding, and an α of 1.88. The index profile was evaluated for $\lambda = 0.6328\text{ }\mu\text{m}$, which is a commonly used wavelength for profiling lightguides. The core and cladding compositions provide a simulated fiber with a N.A. slightly >0.23 and with a core diameter approximately that used for most graded-index multimode lightguides. The refractive index was calculated at $2.5 \times 10^{-2}\text{-}\mu\text{m}$ increments across the core. These values were then compared with the refractive indices at $0.6328\text{ }\mu\text{m}$ for the measured glasses, and in this way the positions of the compositions in the core were obtained. These positions, of course, do not vary with wavelength. Next, the profile shape for the fiber was determined at $1.28\text{ }\mu\text{m}$. This is the wavelength for which the profile having an α of 1.88 should be roughly optimized (see Fig. 9). From this profile the refractive index values at the material positions were found. These values are those which the compositions should have acquired were there constant profile dispersion. These indices were compared with the actual numbers determined from the Sellmeier dispersion relation. The variations in index profile were as follows: 1.8 % GeO_2 , 1.11×10^{-5} ; 2.9 % GeO_2 , 1.53×10^{-4} ; 4.1% GeO_2 , 1.14×10^{-4} ; 7.1% GeO_2 , -2.09×10^{-4} ; 10.8% GeO_2 , -2.61×10^{-4} . One can see from these data that the index profile would deviate from the power-law shape. However, the data show that the deviation in index at any point with this appreciable variation in wavelength is quite small. It is $\sim 3\%$ of the known variation in index in each layer resulting from unwanted changes in composition occurring during fabrication. Additionally, it falls within the standard deviation of many profile measurement schemes.

Using the mixed Sellmeier model the wavelength of zero material dispersion was determined for the entire range of $\text{GeO}_2\text{-SiO}_2$ glasses. The resultant data are compared with dispersion measurements on glasses of various germanium silicate concentrations in Fig. 11. The measured material dispersion was obtained on bulk^{5,14,17,25} and fiber specimens.^{29,30} This data suggest that the actual curve describing the dependence of the zero point on composition might be more convex upward than indicated by the model.

IV. Conclusion

The refractive-index dispersion of GeO_2 glass has been shown to deviate significantly from that of SiO_2 . The calculated material dispersion goes to zero at $1.738\text{ }\mu\text{m}$. A model was presented which is based on the Sellmeier dispersion formula. The model uses a linear variation in its parameters from those for SiO_2 to those for GeO_2 . Without an attempt to minimize residuals, the model fits the data with approximately the same precision as linear regression. Furthermore, it is applicable to obtaining estimates of other dispersion information for the system at any wavelength. Over limited composition ranges the model predicts an approximately linear dependence on composition for n

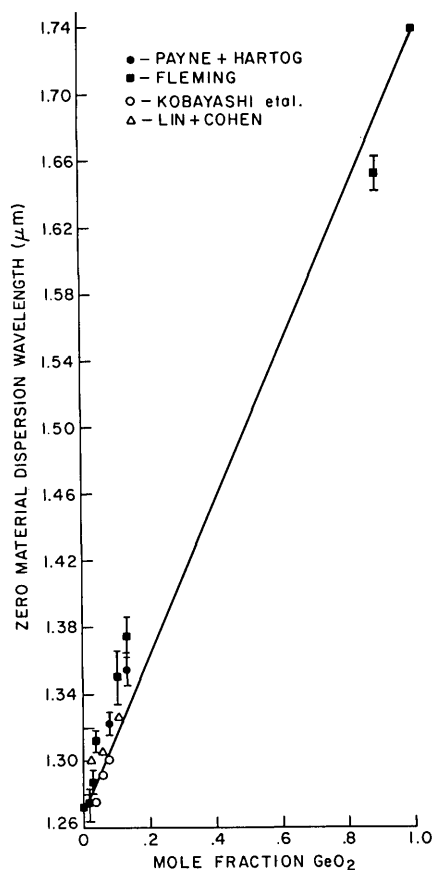


Fig. 11. Wavelength of zero material dispersion for GeO_2 - SiO_2 glasses calculated from the mixed Sellmeier model compared with various measurements on bulk and fiber specimens.

and $dn/d\lambda$. However, it is shown that even the slight nonlinearity is sufficient to cause changes in profile dispersion.

Bulk sample dispersion measurement data were used to show the actual variation in index from that predicted using the no profile dispersion approximation. While these data show statistically significant nonlinearity between $dn/d\lambda$ and n , the resultant variation in index profile is below the resolution of current profile measurement techniques. Additionally, current lightguide fabrication techniques incorporate numerous structural perturbations which cause the fiber core to deviate from any given predicted optimum shape. Therefore, although profile dispersion is present in GeO_2 - SiO_2 fibers, the constant profile dispersion assumption is now considered a reasonable approximation.

The author would like to thank S. R. Nagel, G. E. Peterson, U. C. Paek, and S. H. Wemple for helpful discussions.

References

- R. Olshansky and G. W. Scherer, Optical Communications Conference, Amsterdam, Sept. 1979, Abstract 1.103.
- K. Nassau, "The Material Dispersion Zero in Infrared Optical Waveguide Materials," *Bell Syst. Tech. J.* **60**, 237 (1981).
- H. Osanai *et al.*, "Effect of Dopants on Transmission Loss of Low-OH-Content Optical Fibers," *Electron. Lett.* **12**, 549 (1976).
- S. H. Wemple, "Material Dispersion in Optical Fibers," *Appl. Opt.* **18**, 31 (1979).
- D. L. Wood and J. W. Fleming, "Computerized Refractive Index Measurements for Bulk Materials at UV, Visible, and IR Wavelengths," *Rev. Sci. Instrum.* **53**, 1 (1982).
- W. A. Burnette and C. S. Roberts, AT&T Bell Laboratories, Murray Hill, NJ., private communication.
- L. M. Dennis and A. W. Laubengayer, "Germanium XVII. Fused Germanium Dioxide and Some Germanium Glasses," *J. Phys. Chem.* **30**, 1510 (1926).
- V. A. Kolesova and E. S. Sher, "Two-Component Glasses of the System GeO_2 - SiO_2 ," *Inorg. Mat.* **9**, 1018 (1973).
- Y. Y. Huang, A. Sarkar, and P. C. Schultz, "Relationship Between Composition, Density, and Refractive Index for Germania Silica Glasses," *J. Non-Cryst. Solids* **27**, 29 (1978).
- C. R. Kurkjian and R. W. Douglas, "The Viscosity of Glasses in the System Na_2O - GeO_2 ," *Phys. Chem. Glasses* **1**, 19 (1960).
- G. G. Devyatikh *et al.*, "Material Dispersion and Rayleigh Scattering in Glassy Germanium Dioxide, a Substance with Promising Applications in Low-Loss Optical Fiber Waveguides," *Sov. J. Quantum Electron.* **10**, 900 (1980).
- S. H. Wemple *et al.*, "Binary SiO_2 - B_2O_3 Glass System: Refractive Index Behavior and Energy Gap Considerations," *J. Appl. Phys.* **44**, 5432 (1973).
- C. R. Hammond, "Silica-Based Binary Glass Systems: Wavelength Dispersive Properties and Composition in Optical Fibers," *Opt. Quantum Electron.* **10**, 163 (1978).
- S. Kobayashi, S. Shibata, and T. Izawa, "Refractive-Index Dispersion of Doped Fused Silica," in *International Conference on Integrated Optics and Optical Fiber Communication*, Tokyo, (1977), p. 309.
- R. D. Maurer and P. C. Schultz, U.S. Patent 3,844,550 (1975).
- J. W. Fleming and E. A. Sigety, "Properties of Glasses in the GeO_2 - B_2O_3 - SiO_2 System," *Abstract, Am. Ceram. Soc. Bull.* **53**, 664 (1974).
- J. W. Fleming, "Material and Mode Dispersion in GeO_2 - B_2O_3 - SiO_2 Glasses," *J. Am. Ceram. Soc.* **59**, 503 (1976).
- J. W. Fleming and J. R. Simpson, previously unpublished.
- I. H. Malitson, "Refractive Index of Fused Silica," *J. Opt. Soc. Am.* **55**, 1205 (1965).
- R. Olshansky and D. B. Keck, "Pulse Broadening in Graded-Index Optical Fibers," *Appl. Opt.* **15**, 483 (1976).
- J. A. Arnaud and J. W. Fleming, "Pulse Broadening in Multimode Optical Fibers With Large An/n : Numerical Results," *Electron. Lett.* **12**, 7 (1976).
- H. M. Presby and I. P. Kaminow, "Binary Silica Optical Fibers: Refractive Index and Profile Dispersion Measurements," *Appl. Opt.* **15**, 3029 (1976).
- F. M. E. Sladen, D. N. Payne, and M. J. Adams, "Definitive Profile-Dispersion Data for Germania-Doped Silica Fibers Over an Extended Wavelength Range," *Electron. Lett.* **15**, 469 (1979).
- M. G. Kendall and A. Stuart, *Advanced Theory in Statistics V. Distribution Theory* (Hafner, New York, 1961).
- J. W. Fleming, "Material Dispersion in Lightguide Glasses," *Electron. Lett.* **14**, 326 (1978).
- G. E. Peterson, A. Carneval, U. C. Paek, and J. W. Fleming, "Numerical Calculation of Optimum α for a Germania-Doped Silica Lightguide," *Bell Syst. Tech. J.* **60**, 455 (1981).
- N. Shibata, S. Shibata, and T. Edaishi, "Refractive Index Dispersion of Lightguide Glasses at High Temperature," *Elect. Lett.* **17**, 310 (1981).
- W. A. Sproson, K. B. Lyons, and J. W. Fleming, "Nondestructive Concentration Profiling of Fiber Optic Preforms by Analysis of Raman Spectra," *J. Non-Cryst. Solids* **45**, 69 (1981).
- D. N. Payne and A. H. Hartog, "Determination of the Wavelength of Zero Material Dispersion in Optical Fibers by Pulse-Delay Measurements," *Electron. Lett.* **13**, 627 (1977).
- L. G. Cohen and C. Lin, "Phase Delay Measurements in the Zero Material Dispersion Wavelength Region for Optical Fibers," *Appl. Opt.* **16**, 3136 (1977).